TESTING THE REALITY OF FOCUS DOMAINS

CARLOS GUSSENHOVEN
Katholieke Universiteit Nijmegen

An experiment was carried out which was aimed at testing the hypothesis that, in English, prominence on a predicate contains no information about the "given-new" status of that predicate if it is followed by a (lexical) object that carries a sentence accent. The results are taken to lend support to a view of English prosody whereby (a) prosodic structure is not (fully) predictable from syntactic structure, and (b) the relation between "new"-ness and a sentence accent is indirect.

INTRODUCTION

From the model for the assignment of sentence accents described in Gussenhoven (in press), a number of testable hypotheses can be derived. One of the more important of these concerns the phonological status of pre-final (or pre-nuclear) accents.[1] In order to fully appreciate both the genesis and the claims of the hypothesis it is necessary to outline certain features of the model here.

One of the assumptions in the model is that a sentence accent is the surface manifestation of the speaker's employment of the linguistic category "focus": sentence accents signal the focus distribution of sentences, the division of the semantic material into [+focus] (or "new") and [–focus] (or "given") material (cf. Halliday, 1967). Crucially, the two concepts "focus" and "sentence accent" are only relatable through accent assignment rules, in much the same way that, say, [+passive] and the occurrence of passive verb forms are related to each other. In our case, accent assignment rules take focus markings as input and give surface structures with sentence accents on particular words as output. The relation between [+focus] and a sentence accent is thus indirect, and is determined by the form of the sentence accent assignment rule that happens to be applicable. There is, therefore, no a priori reason why, either, a word that is [+focus] should be accented, or, why an accented word should be [+focus]. It is the former circumstance that will appear to be relevant to our hypothesis.

[1] *I am indebted to Gill Brown for her suggestion to use context retrievability as a means of establishing the linguistic structure of utterances. I thank Mr. A. Dunbar of Queen Margaret's College, Edinburgh, for arranging the recording sessions, and his students and Janet Mackenzie for doing the recordings. I thank Anne Anderson for her advice on the statistical processing of the results. The BMDP programs were developed at the National Health Sciences Computing Facility, UCLA, and sponsored by NIH Special Research Resources Grant RR-3. The date of the latest revision of the program that was used (PV2) is November 1979. I should like to thank Anne Anderson, Ton Broeders, Gill Brown, Anne Cutler and Bob Ladd for their comments on earlier versions of this report.*

The main rule in our model is the Sentence Accent Assignment Rule, or SAAR. SAAR is applicable if at least one complete major semantic constituent is [+focus]. Thus, not counting partial marking of constituents, major semantic constituents are assumed to be either [–focus] or [+focus]. There are three major semantic constituents: (1) Arguments, which figure in sentences as subjects and objects (As); (2) Predicates, which figure as syntactic predicates (Ps); and (3) Conditions, which figure as adverbials (Cs). Minimally, a complete sentence consists of an A and a P. Maximally, leaving embedded sentences out of account, there is only one P, there may be up to three As, and there may be any number of Cs. In (1), there are two As (*I* and *Jane*), a P (*met*) and a C (*in London*); in (2), there is one A (*the television*) and a P (*has gone on the blink*).

(1) I met Jane in London

(2) The television has gone on the blink

The minimum sentence structure, i.e. an A and a P, is at the same time the maximum focus domain. A focus domain is any structure that can be marked as being entirely [+focus] with only a single accent. In the maximum focus domain (AP or PA), SAAR assigns the accent to the A.[2] Any additional constituents form their own focus domain, and are therefore assigned an accent (always assuming they are [+focus]). Put differently, all [+focus] constituents are assigned an accent by SAAR, except a [+focus] P that has merged with a [+focus] A into a single focus domain. SAAR thus divides the sentence into focus domains and gives every domain an accent. The rule could be formulated as in (3), where underlining stands for [+focus] and absence of underlining for [–focus], and where | is a focus domain boundary, and X stands for any constituent.

(3) SAAR:

Domain assignment	$\underline{P}(X)\underline{A} \rightarrow \mid \underline{P}(X)\underline{A} \mid$
	$\underline{A}(X)\underline{P} \rightarrow \mid \underline{A}(X)\underline{P} \mid$
	$\underline{X} \rightarrow \mid \underline{X} \mid$
Accent assignment	$\mid \; \mid \rightarrow \mid * \mid$ (in AP/PA, * goes to A)

To illustrate, if in (1) everything except the subject is [+focus], SAAR applies as illustrated in (4). If also *Jane* is [–focus], it applies as illustrated in (5). And if all of (2) is [+focus], SAAR produces (6).

(4)

Focus distribution	$A\ \underline{P\ A\ C}$
Domain assignment	$A \mid \underline{P\ A} \mid \underline{C} \mid$
Accent assignment	$A\ P\ \overset{*}{A}\ \overset{*}{C}$ (I met J$\overset{*}{\text{a}}$ne in L$\overset{*}{\text{o}}$ndon)

[2] *Linguists in fact frequently display the effect of the rule when discussing the syntactic concept of object incorporation, glossing such instances as, for example,* I FISH-like, He BEER-drinks, *rather than* *I fish-LIKE, *He beer-DRINKS. *Note that it is not suggested that it is impossible for both the Argument and the Predicate in a PA combination to be assigned an accent, as in You EAT FISH, but you DRINK BEER.*

(5) Focus distribution $\quad A\ \underline{P}\ A\ \underline{C}$

Domain assignment $\quad A \mid \underline{P} \mid A \mid \underline{C} \mid$

Accent assignment $\quad A\ \overset{*}{P}\ A\ \overset{*}{C}$ (I m$\overset{*}{\text{e}}$t Jane in L$\overset{*}{\text{o}}$ndon)

(6) Focus distribution $\quad \underline{A\ P}$

Domain assignment $\quad \mid \underline{A\ P} \mid$

Accent assignment $\quad \overset{*}{A}\ P$ (The t$\overset{*}{\text{e}}$levision has gone on the blink)

There are a number of constraints on the construction of maximum focus domains. That is, not all As are capable of merging with Ps, and not all Ps are capable of being merged with As. In such cases, As and Ps are separated by a focus boundary, and – if both are [+focus] – SAAR applies so as to put an accent on each of them. One type of A that cannot merge with Ps is the class of independent quantifier, containing items like *nothing, no one, everyone.* Thus, while we can have an AP focus domain in (7), we cannot have one in (8):

(7) A: What happened? B: The pr$\overset{*}{\text{i}}$soners have escaped

(8) A: What happened? B: *$\overset{*}{\text{E}}$verybody has escaped

The correct structure for B's reply in (8) being $\underline{A}$ (quantifier) $\underline{P}$, giving $\mid \underline{A} \mid \underline{P} \mid$, or *$\overset{*}{\text{E}}$verybody has esc$\overset{*}{\text{a}}$ped.*

A Consequence of the Model

While the above remarks are necessarily sketchy, they suffice to enable the following consequences to be derived:

(9) 1. A sentence with a regularly merging PA combination in final position with an accent on the A, is ambiguous between a structure in which the P is [+focus] and one in which the P is [–focus], since in either case SAAR will assign an accent to the A: in the version in which the P is [–focus], the [+focus] A forms a focus domain by itself, and in the version in which the P is [+focus], it shares its focus domain with the [+focus] A, which will get assigned the accent.

2. A sentence with, for example, a PC sequence in final position in which the C is accented, and where the P is [+focus], is phonetically distinct from an otherwise identical sentence in which the P is [–focus]: if the P is [–focus], no accent is assigned to it, but if it is [+focus], it is assigned an accent, as it has a focus domain to itself.

Symbolically, the situation can be represented as in (10),

(10)

	a	b	c	d
Focus distribution	$\underline{PA}$	$P\underline{A}$	$\underline{PC}$	$P\underline{C}$
Domain assignment	$\vert \underline{PA} \vert$	$P \vert \underline{A} \vert$	$\vert \underline{P} \vert \underline{C} \vert$	$P \vert \underline{C} \vert$
Accent assignment	$P\overset{*}{A}$	$P\overset{*}{A}$	$\overset{*}{P}\overset{*}{C}$	$P\overset{*}{C}$

where the surface forms in (a) and (b) are identical, but those of (c) and (d) are not. In terms of a concrete example, this would mean that if to the question *What does he do?*, the answer is given *He teaches linguistics*, then this answer could equally well have served as an answer to the question *What does he teach?* ((a) and (b) in (10)). In either case, SAAR assigns an accent to *linguistics*, and the fact that *teaches* is [+focus] in one case and [–focus] in another has no effect on the surface forms. But if to the question *What does he do?* the answer is given *He teaches in Ghana*, then this latter utterance cannot serve as a well-formed answer to the question *Where does he teach?*, since in the former situation the P *teaches* and the C *in Ghana* form separate focus domains and are each assigned an accent, while in the latter only the [+focus] *in Ghana* is assigned an accent ((c) and (d)).

Investigation of the literature shows that the issue that this prediction raises, has in fact been discussed, though usually only cursorily, and no unambiguous answer can be established. Thus, Halliday (1967, p. 208) claims that a pronunciation of *They teach classics* with *teach* "new," is rhythmically distinct from one with *teach* "given." Postal (1971, p. 234) claims that *Charlie insulted his father* with "contrastive stress" on *father* is phonetically distinct from a version with "ordinary stress throughout." Palmer (1974, pp. 222-3) claims that *This student wants to drop linGUIStics* can be equivalent both to *This student wants to give linGUIStics up* and to *This student wants to give linguistics UP*, implying that *drop* can be unaccented as well as accented in the way illustrated in the two putative paraphrases. Enkvist (1979) claims that *Did Charlie kick the bucket?* must have an accent on *kick* in addition to one on *bucket* if *kick the bucket* is to be understood as 'die'; if only *bucket* has an accent, so he says, the sentence will be given its literal interpretation. Chafe (1970, p. 225), by contrast, says that *David emptied the BOX* is ambiguous between a reading with "both the verb root and the patient noun [as] new" and one with only the patient noun as new, quite in line with the first consequence of our model. And, notwithstanding their disagreement over other aspects of sentence accents, Chafe's position would be supported by both Chomsky (1972, p. 203) and Bing (1979, p. 180). In none of these cases was experimental evidence provided. Neither did we find an indication that not all PA sequences can be treated alike.

A hypothesis

If the consequences in (10) can be shown to hold good in experimental conditions, then, obviously, this would constitute important evidence in favour of our model, and more generally, against any model that relies on syntactic considerations alone. Subsequent to the derivation of the consequences in (9), therefore, a testable hypothesis needed to be formulated. To this end, 32 dyads were constructed, each consisting of a question, henceforth the context, and an answer, henceforth an utterance. The utterances in one half of the dyads had a merging PA combination in final position. This group of 16 dyads is referred to as Structure A. The utterances in the other half of the dyads contained various combinations of semantic constituents in final position (PC, AC, a non-merging PA) both of which would be assigned a SAAR accent if they are [+focus]. This group of 16 dyads is referred to as Structure B. In each Structure one half of the dyads comprised contexts that may be expected to force the corresponding utterances to have a [+focus] pre-final constituent followed by a [+focus] final constituent, while the other half comprised contexts that would produce identical utterances, except for the fact that the pre-final constituent is [–focus]. These two groups of dyads are referred to as the [+focus] and the [–focus] groups, respectively. Crossing the two variables gives us four sets of eight dyads. The following are representative examples in each set:

(11)	Structure A, [+focus]	C: Do you live by yourself?
		U: I share a flat
	[–focus]	C: I hate sharing things, don't you?
		U: I share a flat
	Structure B, [+ focus]	C: Please tell me what happened that night
		U: I remember nothing
	[–focus]	C: What do you remember from your last lesson?
		U: I remember nothing

Our model predicts that if in taped recordings of these dyads we interchanged each [+focus] utterance with its [–focus] counterpart, listeners will be unable to tell the spliced dyads from the original ones in the case of Structure A, but, in the case of Structure B, ought to be able to tell the spliced dyads and the original dyads apart with a certain measure of success. We will refer to this measure of success as the Context Retrievability Score, or CRS, to be expressed as a percentage. This provides us with the following testable hypothesis:

(12) Utterances in Structure A have a chance (50%) CRS, while utterances in Structure B have a CRS of between 50% and 100%.

Observe that the CRS in Structure B will depend on the extent to which the presence or absence of the pre-final accent can be perceived. It is reasonable to assume that the ease with which this can be done is related to the distance between the accentable syllable of the pre-final constituent and the accentable syllable of the final constituent. If no unaccented syllables intervene, the difference between presence and absence of the accent on the pre-final constituent will be more difficult to establish for a listener than when some three or four unaccented syllables intervene. In this latter case, the pre-final accent will have more scope to stand out. For this reason, the utterances in Structure A and Structure B were matched for the number of unaccented syllables occurring between the accentable syllable of the pre-final constituent and the accented syllable of the final constituent. Four out of the 16 dyads in each group had utterances with zero intervening syllables, while there were four with one, four with two and four with three intervening syllables in their utterances. A second clause should therefore be added to our hypothesis:

(12′)

1. (= 12)
2. Distance between potential accents will have no effect on the CRS in Structure A, but a graded positive effect in the case of Structure B.

It was thought desirable to define a practical upper limit of perceivability. Maximum perceivability of an accent was taken to be achieved if instead of occurring before another accent, the potential accent was itself final. This would allow the presence or absence of an accent on that syllable to be perceived as the occurrence of what is frequently called "normal" nucleus placement versus "contrastive" nucleus placement. To this end, a third group of 16 dyads was constructed, all of them having an Argument in final position. Contexts were chosen so as to force the Argument in one half of the utterances to be [+focus], and in the other half to be [–focus], these two groups of utterances being otherwise identical. The effect, of course, is that the [+focus] utterances, but not the [–focus] ones, will have a sentence accent on the final word. This group of 16 dyads is referred to as Structure C. A representative example of a [+focus] dyad in this group and its [–focus] counterpart is given in (13). A complete list of all the dyads in all groups is given in the Appendix.

(13) Structure C

[+focus]	C:	Anything in particular you want me to say to her?
	U:	Tell her I'm planning a coup
[–focus]	C:	She seemed sort of sorry there'd never be another coup
	U:	Tell her I'm planning a coup

If contexts and utterances are switched as in Structures A and B, we would expect the spliced dyads to be perceived as grossly ill-formed in comparison to the original dyads. In the dyads in Structure C, distance between potential accents was varied: taken in sets of four dyads, utterances varied from zero intervening words, via one and two, to as many as seven and 12. For obvious reasons, this distance was not expected to have an effect on the CRS. We revise our hypothesis as follows:

(12″)

1. Utterances in Structure A have a chance CRS, utterances in Structure C have a CRS approaching 100%, while utterances in Structure B will have a CRS between those for Structures A and C.
2. Only in Structure B will the distance between potential accents have an effect on the CRS. This will take the form of a positive correlation.

As far as could be established, the literature provides no indication that (12″) is a correct prediction. Thus, irrespective of whether the answer to *What does he do?* is *He teaches linguistics* or *He teaches in Ghana,* the British literature would consider *teaches* as "stressed" in either case. Similarly, a SPE description (Chomsky and Halle, 1968, p. 18) would assign a [2stress] to the first syllable of *teaches* in either case, and neither would any version of metrical phonology (Liberman and Prince, 1977; Selkirk, 1980) discriminate between the two stresses. This suggests that our hypothesis is not one that readily emerges from the exercise of any form of linguistic intuition. Indeed, informal interviews showed that native speakers do not consciously regard the prosodic statuses of the two instances of *teaches* above to be different, and consider both "stressed." Now, our hypothesis does not rule out that at the position of the P in a Structure-A utterance, prosodic options can be employed that are unrelated to the option [±focus], but which may – though need not – be realized by some use of fundamental frequency (F_0) or of other phonetic variables that is different from the way focus-expressing accents are realized. What our hypothesis does claim is that at the position of the P in Structure-A utterances, no prosodic features will be found that realize an accent of the sort here represented as *. Specifically, the hypothesis does not rule out the use of high onset at *teach-* in *He teaches linguistics.* Again, what it does claim is that this high onset will not be taken by listeners to signal that *teaches* is [+focus], or rather that it will be taken to be irrelevant to focus.

What this would mean is that when asked to give "stress" judgments, listeners will respond non-differentially to "phonetic prominence," and will not allow their judgments to be guided by the phonological source of that prominence. Hence, if prediction (10) is correct, then the presence or absence of prominence-lending features on the P in Structure-A utterances should not have an influence on the success with which listeners are able to match utterances and contexts. By contrast, in the case of Structure-B utterances, there should be a positive correlation between the degree of perceived stress on the pre-final constituent and the extent to which listeners are prepared to consider the utterance a [+focus] one, since it was claimed earlier that in this group the ease with which the pre-final accent could be perceived was going to be related to the CRS.

We should therefore revise our hypothesis as follows:

(12‴)

1. (see (12″))
2. (see (12″))
3. The CRS of the utterances in Structure B, but not that of the utterances in Structure A, correlates positively with the perceived stress on the pre-final constituent.

Note, incidentally, that although the correlation in Structure C will, predictably, be positive, no theoretical interest is at stake here, and the prediction should not figure in the hypothesis.

In order to test (12‴), two experiments were carried out:

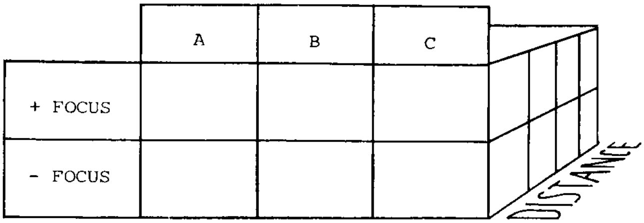


Fig. 1. Design of the Context Switching Experiment.

1. A context-switching experiment
2. A stress perception experiment

The first test was needed to test the first and the second clause of the hypothesis (see (12″)), and requires the design shown in Figure 1. The second test was needed to test the third clause of (12‴). This is a straightforward test whose results are to be correlated with the results of the first test. The prediction is that there is a positive correlation for Structure B and no correlation for Structure A.

The tests are described, and their results given, in separate sections. In a final section some implications are discussed.

THE CONTEXT-SWITCHING EXPERIMENT

In order to obtain the stimuli required for the Context Switching Test, six native speakers of Standard British English were recruited. Four of them were slightly Scottish accented. They were variously combined in pairs. Pairs of speakers were seated comfortably on opposite sides of a table in a spacious studio, and were provided with scripts containing the 48 dyads to be used in the experiment. Dyads containing lexically identical utterances were spread as wide as possible in the script. One speaker in each pair then read out the context, and the other the utterance of each dyad, which speech was recorded on magnetic tape. The exercise proved more difficult than was expected. Although five of the six speakers were students at a Drama College, speakers frequently read out the utterances in a mode that Brazil, Coulthard and Johns (1980, p. 83) refer to as reading "what it says," putting stresses on all major words, quite regardless of context. The problem seemed most acute in Structure C, where the utterances that were intended as [–focus] were often given an accent on the last word. Also, some speakers tended to overact. These problems, which were no doubt partly due to the fact that speakers were unaware of the purpose of the sessions, could up to a point be solved by asking the speaker of the utterances to look up from the script when listening to the context and giving the utterance. In addition, speakers were frequently asked to do certain dyads again. These sessions produced some two hours' worth of recordings.

From these recordings one complete set of 48 dyads was selected that (a) were not obviously instances of "overacting" or reading "what it says," (b) had a reasonable spread over the different speakers, and (c) if belonging to Structure A, did not display a clear step-up in pitch on [–focus] Ps, or clearly lack a step-up in pitch on [+focus] Ps. This third criterion ensured that it was possible to test the third clause in hypothesis (12'''). Clearly, if we are to be able to demonstrate that stress differences on the Ps in Structure A are not related to the perception of focus, we must make sure that these stress differences are there to begin with. In addition, a set of 48 contexts was selected.

Composition of the test tape

A test tape was prepared, in which the 48 contexts were combined both with the 48 corresponding utterances and with the 48 utterances that had the opposite focus value for the constituent concerned. These 48 pairs of dyads thus consisted of two identical occurrences of a context, one followed by a [+focus] utterance, and the other by the lexically identical [–focus] utterance. The following criteria were applied when compiling dyads and test tape:

1. The two utterances in each pair of dyads were spoken by the same speaker. This speaker was different from the speaker of the context.

2. In no case was the utterance actually spoken in the context it combined with on the test tape: the original pairs of speakers were split up and crossed. This was done in order to avoid tessitura-matching effects. Brown, Currie and Kenworthy (1980, p. 123) found that speakers tended to match the tessitura or "key" of their utterance with that of the last utterance of their interlocutor, which tendency could well have had a biasing effect on our scores.

3. If a pair of utterances appeared in one order in the [+focus] context, it appeared in the same order in the [–focus] context. This was done in order to cancel out possible order effects of the sort reported in Gussenhoven and Blom (1978). They found that for the first of a pair of stimuli, a perceptually small amount of stimulus attribute is overrated by listeners, and a perceptually large amount is underrated. The effect is explained as resulting from the judges' concern that the small amount of the first stimulus is going to be swamped by the expected larger amount of the second stimulus, which concern then causes the overestimation. Conversely, a large amount might induce subjects to feel over-confident about having mentally registered it, causing them to underestimate it when the second stimulus comes along.

4. If a pair of lexically identical utterances appeared in one order (see 3. above), then the pair that had the same specification for Structure and distance between potential stresses, appeared in the opposite order.

5. Clusterings of speakers and utterance-types were avoided.

Stimulus-to-stimulus periods varied with the duration of the dyads, and averaged approximately 18 secs. The duration of the test tape, including trial items, was 16 minutes.

Presentation of the test tape

The test was presented to a group of 24 judges in a language laboratory provided with headphones. The judges, all native speakers of English, were recruited on a voluntary, no-fee basis from groups of first- and second year students at the University of Edinburgh, attending classes a few days before the test was given. In the instruction, they were told that they would hear a series of question-and-answer combinations, in which one combination was actually spoken as such, and the other was made up of the same question and an answer that was spoken in response to some other question, and that if they listened carefully, it was always possible to tell which was the correct combination. It was pointed out to them that in both the correct and the incorrect combination, they would hear traces of editing. On their score sheet, the context was given by way of stimulus identification. They were asked to register their judgment by writing either "1" or "2" in a box provided for each stimulus.

Results

A score of 100% for a group corresponds to a CRS of 16 (utterances) times 24 (judges), or 384. Figure 2 gives the overall CRS per Structure in percentages. As predicted, the CRS in Structure B is poised between those of Structure A and Structure C. A three-way analysis of variance was performed on the data (Dixon and Brown, 1979). The dependent variables were **Structures** (Structures A, B and C), **Distance** (distance between potential stresses: four levels, only Structures A and B being precisely matched), and **Focus** ([+focus] and [–focus] groups). There was a significant main effect for **Structures** ($F = 38.67$, $p < 0.0001$), and there were significant interaction effects of **Distance** by **Structures** ($F = 4.26$, $p < 0.001$) and **Focus** by **Structures** ($F = 5.77$, $p < 0.01$). The scores of Structure C differed both from those in Structure A and from those in Structure B at $p < 0.001$, according to a post-hoc Scheffé-test. As our interest in Structure C was really confined to its status as an indication of the maximum CRS that is obtainable in practice, scores in this group should really not be allowed to affect the significance of the differences between the CRS in Structure A and that in Structure B. Therefore, a second analysis of variance was performed on the data in Structures A and B only. Table 1 gives the results.

TABLE 1

Three-way analysis of variance of the scores in Structures A and B

	*df*1	*df*2	*F*	*p*
Structures	1	23	14.4	< 0.001
Distance	3	69	2.2	n.s.
Focus	1	23	0.35	n.s.
Interaction Structures-Distance	3	69	4.45	< 0.01
Interaction Structures-Focus	1	23	9.36	< 0.01

Observe that the significant effects noted earlier are retained in this analysis. The significant effect for **Structures** indicates very clearly that a null hypothesis that the utterances in structure B were drawn from the same population as those in Structure A must be rejected at a very high level of significance. It thus supports the claim made in the first clause of (12‴).

The interaction of the distance between potential prominence peaks with the Structure to which the utterance belongs has been plotted in Figure 3, in which also the C scores have been included for comparison. The important point to notice is that **Distance** does not improve the CRS in Structure A, but, with the exception of the category "one syllable intervening between potential stresses," does do so in Structure B. The irrelevance of pre-nuclear prominence on the P of the utterances in Structure A to the perception of focus is thereby further underlined. These results support the second clause of hypothesis (12‴).

The significant interaction between the focus value of the pre-final constituent and the Structure to which the utterance belongs was unexpected. From the scores it appears that, on the whole, the [–focus] utterances in Structure A did better than the [+focus] utterances (with scores of 115 and 91 respectively), while in Structure B the opposite tendency could be detected (with scores of 124 and 139 respectively). That is, in Structure A, but not in Structure B, judges were somewhat more successful in matching the [–focus] context with the [–focus] utterance than in matching the [+focus] context with the [+focus] utterance. Since we have no theory as to how pre-nuclear stress on the predicate in Structure A could be relevant to the CRS, and since, moreover, within either Structure A or Structure B the differences do not even begin to approach a significant level, we will not further investigate these results here.

Since each "cell" formed by **Structure, Distance** and **Focus** contained two pairs of lexically identical utterances, we were able to split the materials into two halves, and treat these as the experiment and its replication. An analysis of variance was performed on the data in Structures A and B with a grouping variable representing these two halves

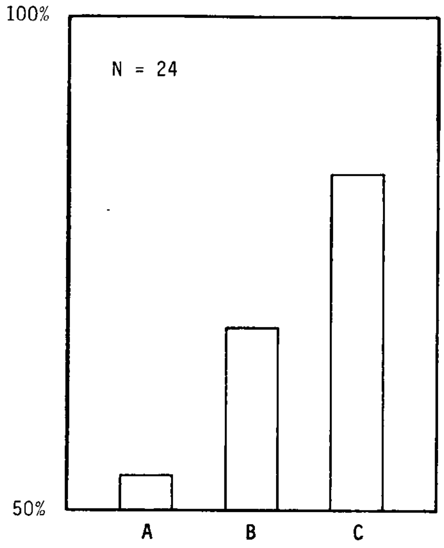


Fig. 2. Overall CRS (%) in Structures A, B and C.

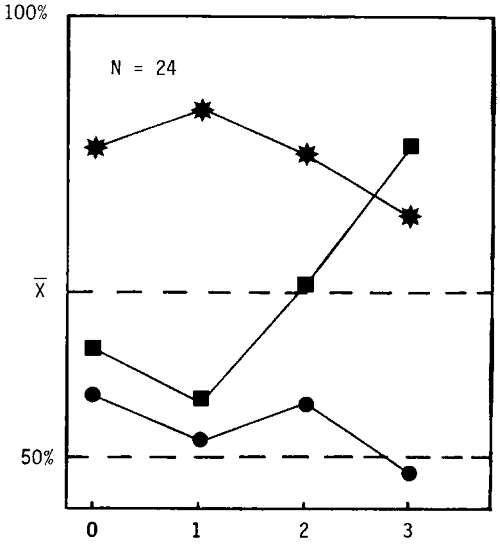


Fig. 3. CRS as a function of the distance between potential accents in Structure B (● =Structure A, ■ =Structure B, ✸ =Structure C; for 0, 1, 2, 3 see text).

of the data. There was no significant effect for this grouping variable ($F = 0.01, p = 0.94$). Moreover, when each half was analysed separately, differences between Structure A and Structure B remained significant in both halves ($p < 0.01$ in one case and $p < 0.05$ in another). The interaction between **Distance** and **Structures** remained significant in one half at $p < 0.05$, while there was a trend in the other half for this interaction. Thus, for two different groups of stimuli, the same pattern of results was found. This strongly suggests that the effects observed do not result from the particular stimuli used in each condition, but reflect more general differences between the structures listed under (9).

THE STRESS JUDGMENT TEST

In order to adduce evidence bearing on the third clause of hypothesis (12′′′), a Stress Judgment Test was carried out. An additional reason for carrying out this test was that the material used in the Context Switching Experiment had been (non-randomly) selected. It is therefore conceivable that differences in stress value between the [–focus] and [+focus] utterances in Structure A were very much smaller than those in Structure B, and that a potential cue to context in Structure A might thus have been effectively removed. In spite of the care that was taken when selecting the utterances, we needed to be sure that this was not the case.

The 48 utterances that were used in the Context Switching Experiment were randomized, and a new test tape was prepared. Stimuli consisted of two occurrences of each utterance, and were approximately 14 seconds apart, the total test duration being about 12 minutes. The test was presented to a group of 14 native speakers of British English. They were recruited in the same way as the judges who took part in the Context Switching Experiment, and took part on the same basis. None of them had also taken part in the Context Switching Experiment. The test was presented to them in a language laboratory equipped with headphones. Judges were instructed to rate the degree of stress on the experimental syllable of each utterance (i.e. the pre-final accentable syllables in Structures A and B, and the final accentable syllable in Structure C), on a scale from 1 (for "no stress") to 5 (for "maximum stress"). On their score sheets, each utterance was given, with a large box over the syllable to be rated for stress. Subjects did not appear to find the task difficult.[3]

Results

Mean scores for the [–focus] and [+focus] groups in each of the Structures A, B and C are given in Figure 4. Observe that although the utterances in Structure B were some-

[3] *It might have been expected that it would be easier to rate SAAR accents for stress (as in Structures B and C) than "onset prominence" (as in Structure A). Or that prenuclear prominences (as in Structures A and B) are more difficult to rate than nuclear prominence (as in Structure C). This might have shown up in differences between standard population variances in the various groups. This turned out not to be the case (A: 0.98, B: 0.95 and C: 0.94). Neither was absence of stress more difficult to rate than presence of stress ([–focus] group: 0.93, [+focus] group: 0.97).*

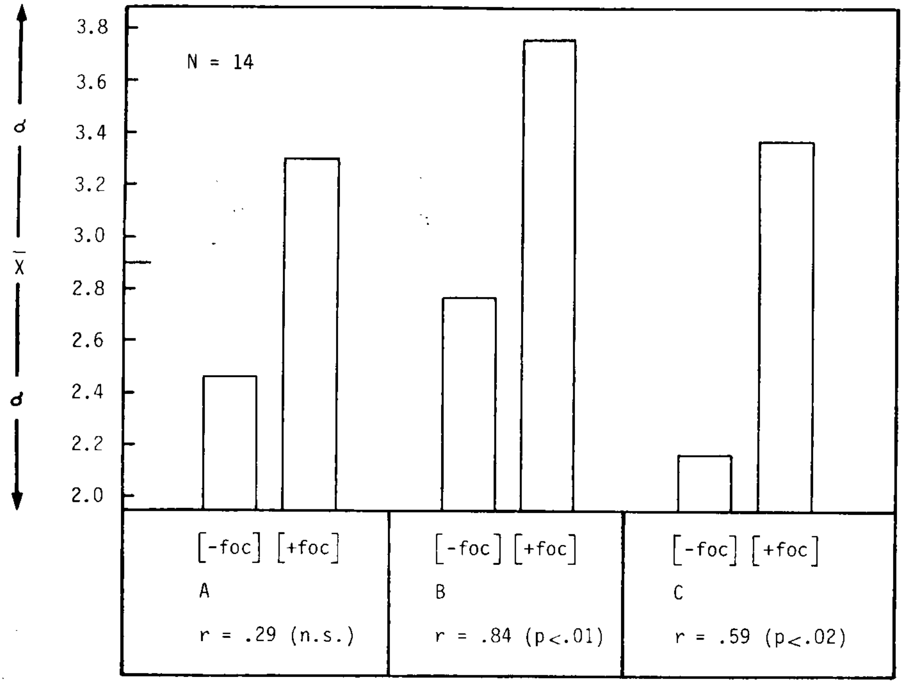


Fig. 4. Observed "stress" levels for the [–focus] and [+focus] utterances in Structures A, B and C, and the product-moment correlations between the observed "stress" levels and CRS for each group.

what more favourably discriminable than those in Structure A, in all three structures there were considerable differences between the subjective stress values of [+focus] and [–focus] utterances. Only in one case in Structure A and in one case in Structure B, did stress judgments turn out to be the reverse of what might have been expected: i.e. the [+focus] utterance of a pair was judged to have marginally less stress (–0.14 in either case) on its experimental syllable than the [–focus] utterance. In both cases this concerned a pair with "one syllable between potential stresses," which would appear to provide a possible explanation for the dip in the B-curve in Figure 3.

Product-moment correlations between CRS and stress judgments were calculated (see Figure 4). As predicted, a high correlation of $r = 0.84$ between the CRS and the independently obtained stress judgments was observed for Structure B. The difference between the correlation coefficients obtained for Structures A and B was significant at $p < 0.02$. The correlation coefficient obtained for Structure A is itself not significant. On the basis of these results, a null hypothesis that there is no difference between Structure A and Structure B in the way their CRS correlates with the perceived stress on the predicates, must be rejected. The results strongly suggest that in the Context

Switching Experiment judges responded categorically differently to the stimuli in Structure A and those in Structure B.

The differences between the correlation coefficient obtained for Structure C and each of the other two correlation coefficients are not significant. While the finding has no bearing on our hypothesis, it may be observed that the intermediate value of the correlation coefficient observed in Structure C may be a reflection of the fact that presence *vs.* absence of a nuclear accent on the final word will not be perceived as a continuum, but categorically. It is either there or it is not there, but whether it is unobtrusively or emphatically there, is unlikely to influence judges' behaviour in a context retrieval task, and will thus not be reflected in the correlation coefficient.

DISCUSSION

The results of the experiment confirm the assumption underlying the model for sentence accent assignment described in Gussenhoven (in press), in which sentence accents are viewed as morphemes manifesting the focus distribution of sentences. As was stated in the introduction, the relation between sentence accents and "focus" (or "importance") is indirect in the sense that absence or presence of a sentence accent on a word does not in itself provide a label indicating the status of that word. Example (2) showed how a P can be both unaccented and [+focus]. As a further illustration of the indirectness of this link, consider (14), where the words *now definitely* (a [+focus] C) have been moved between the [+focus] A and the [+focus] P:

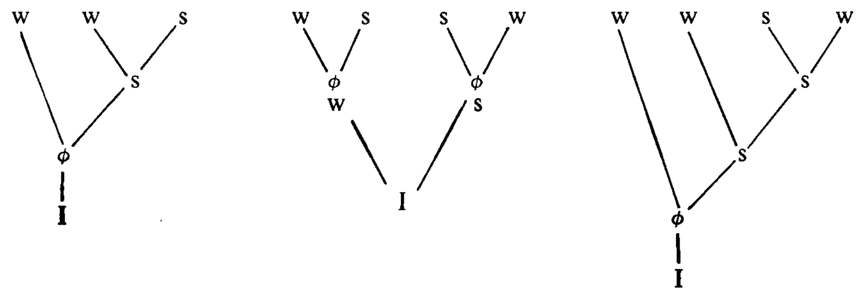
(14) The t*elevision has now d*efinitely gone on the bl*ink

The P receives its sentence accent not because it is now more "important" than in (2), but because it is prevented from merging with the A into a single focus domain by the interposed [+focus] C.

Since focus is a linguistic (semantic) category in its own right, the results support a view of English prosodic structure that is much less dependent on syntactic structure than current descriptions of English prosody suggest. Generally, prosodic structure is believed to be largely derivable from syntactic structure, and much research reflects this in that the aim is to predict the former from the latter. This includes such authoritative accounts of English stress as Chomsky and Halle (1968), Liberman and Prince (1977), and Selkirk (1980). While there is no denying that identical representations of "normal" or "neutral" renderings of *I know John* and *I know no one* capture much of our intuition concerning the surface prosody of such sentences, and no denying even that predicting prosody from syntax/lexis may be an extremely valuable research activity in its own right in an area of applied linguistics (cf. automatic readers for the blind), the point must be made that too much emphasis on the "interpretive" role of the phonological component may cause an important part of the linguistic structure of sentences to be obscured. One condition on any description must surely be that it should distinguish between prominence peaks that result from focus-manifesting accents and those that result from other sources.

Selkirk (1980) in fact provides the elements for a description that is capable of incorporating our findings. She presents a hierarchy of metrically organized phonological units: syllables, feet, phonological words, phonological phrases, intonational phrases and utterances. While a "phonetic packaging" view would seem to be appropriate for the first three types of unit, it is with the introduction of phonological phrases that linguistic facts of the sort the experiment was concerned with begin to fall by the board. Selkirk defines the phonological phrase (ϕ) derivatively, allowing it to coincide with syntactic phrases, like NP, VP, PP. (For details see Selkirk (1980), and for experimental support for her analysis, Nespor and Vogel (in press).) As a result, the difference between merging and non-merging PA combinations, or that between a merging PA combination and a PC sequence, or that between a non-merging PA with a [+focus] P and one with a [−focus] P, all of which must be assumed to be prosodically different, cannot be expressed. If, however, we allowed each ϕ to correspond to the focussed constituent(s) in a single focus domain, together with any associated [−focus] material, then we would have a description that does capture these manifestations of focus distribution. This would produce (15a) as the representation of *I know John* with *John* [+focus] and *know* [+focus] or [−focus]; likewise, (15b) would be the representation of *I know no one* with both *know* and *no one* [+focus], and (15c) that of *I know no one* with only *no one* [+focus]. In (15) I stands for "intonational phrase," as in Selkirk (1980); units above and below ϕ and I have been dispensed with.

(15) a. I know John b. I know no one c. I know no one



It is important to observe that in (15ab) different prosodic structures are postulated for identical syntactic structures, which by any definition are assumed to have "normal intonation." That is, we are not concerned merely – or even mainly – to point out that prosodic phonology should take "contrastive" accentuation into account.

While the main result of the experiment bears on the principled difference between "real" pre-final accents (or *s), and "pseudo" accents, i.e. prominence peaks having a different phonological source, the results also confirm the conclusion reached in Brown, Currie and Kenworthy (1980, p. 115), who point out that there may be more than one accent peak in an utterance, and that a view of the tone group as a single unit of information (e.g., Halliday, 1970, p. 3) is incorrect, at least to the extent that this statement implies that a tone group is characterized by a single, most prominent syllable, the nucleus. That is, [+focus] utterances in Structure B must be assumed to have (at least) two accents that are semantically on a par, both of them signalling the focussed status of the constituents they are found on. This statement has, of course, no bearing on the fact that the last of the SAAR accents will have to be singled out for considerations of "form" (as opposed to "function"). Thus, in Dutch, for example, there exists a rhythm rule that has the tone group as its domain: words that are subject to the rule obligatorily have an unshifted pronunciation when they carry the last SAAR accent of the tone group (e.g. *alTIJD* 'always'), and a shifted one (*ALtijd*) when they occur anywhere before that accent in the same tone group. For this last accent the term "nucleus" could be retained (cf. Gussenhoven, forthcoming).

One of the questions that the results of the experiment raises is whether the judges' differential response to Structures A and B was determined solely by their (tacit) knowledge of focus structure, or that it was at least partly determined by phonetic considerations. Phrased more concretely, is the "stress" that judges claimed to hear on the [+focus] Ps in Structure A phonetically different from the "stress" they claimed to hear on the [+focus] Ps in Structure B? Our stimuli do not provide us with suitable material to answer this question: in practically all cases Ps are segmentally not comparable. Arguably, an experiment with synthetic stimuli would be the best way of determining the answer. Variables to be considered here include duration, size of F_0 step-up on the P, and the occurrence of a valley between this prominence peak and the final accent.

APPENDIX

Below are listed the contexts and utterances in all stimuli in Structures A, B, and C, ordered according to distance between potential accents. Experimental syllables have been italicized, while the last SAAR accent, or nucleus, is capitalized.

Structure A

C: And what is your contribution to society?
C: What exactly is it you are creating?
U: We cre*ate* BUSiness

C: What is the nature of your business?
C: What is it you repair?
U: We re*pair* RADios

C: Do you live by yourself?
C: I hate sharing things, don't you?
U: I *share* a FLAT

C: What are those boys doing in the water?
C: What are they catching?
U: They are *catch*ing FISH

C: What does he do?
C: What does he teach?
U: He *teach*es linGUIStics

C: What did he make that move for?
C: Has he started attacking anything yet?
U: He is at*tack*ing the BISHop

C: What did he say about the invoices this morning?
C: Just what did he promise to do with them?
U: He *pro*mised me to SORT them[4]

C: How did Pat and Vernon spend their time?
C: What were they building there?
U: They were *build*ing a caNOE

Structure B

C: Could it be that you take the wrong view of people in general?
C: Do you distrust me?
U: I dis*trust* NO one

C: I am sorry we cannot employ her
C: Does she sing?
U: She *sings* BEAUtifully

C: Please tell us what happened that night
C: What do you remember from your last lesson?
U: I re*mem*ber NOTHing

C: Why are you looking so worried, dear?
C: Where were they shooting?
U: They were *shoot*ing EVerywhere

[4] *During an informal presentation of this report it turned out that* he promised me to sort them *is not in fact an acceptable sentence to all native speakers, some of whom require finite clause complementation (. . .* me that he would sort them*). We have no reason to suppose that this circumstance will have had an effect on the results of the experiment.*

C: What does he do?
C: Where does he teach?
U: He *teach*es in GHAna

C: Where will you be in January?
C: Where will you be skiing?
U: We will be *ski*ing in SCOTland

C: Do you know anyone in Scotland?
C: Do you know any vicars?
U: I know a *vic*ar in DunDEE

C: What did you think of their house?
C: There were three very nice chimneys . . .
U: I liked the *chim*ney in the DRAWing-room

Structure C

C: What is the best representation of the shortest distance between two points?
C: Now please draw a line
U: A *straight* line? (A straight LINE/ STRAIGHT line)

C: What is the demand this time, Mr. Johnston?
C: I thought you had a canteen
U: We want another can*teen* (another canTEEN/ aNOTHer canteen)

C: Anything in particular you want me to say to her?
C: She seemed sort of sorry there'd never be another coup
U: Tell her I'm planning a *coup* (planning a COUP/ PLANning a coup)

C: Susan's looking after daddy
C: He may be running a temperature
U: Has she taken his *temp*erature? (taken his TEMPerature/ TAKen his temperature)

C: Why doesn't "nuance" count as a word?
C: Have you looked in the dictionary?
U: Sorry, it wasn't in the *dic*tionary (in the DICtionary/ IN the dictionary)

C: Are you sure we're out of sausages?
C: You might have brought me the jars on the shelves
U: There were none on the *shelves* (on the SHELVES/ ON the shelves)

C: They said all sorts of things about her
C: They called her a "bastion of common sense"
U: Surely she was pleased with the fact that they called her a "bastion of common *sense*" (. . . SENSE/ . . . PLEASED . . .)

C: I wonder why he seemed so piqued
C: A "young woolly liberal" is what they described him as
U: He must have been offended by the phrase a "young woolly *lib*eral" (. . . LIBeral/ . . . ofFENded . . .)

REFERENCES

BING, J.M. (1979). *Aspects of English Prosody* (Bloomington, Indiana).
BRAZIL, D., COULTHARD, M. and JOHNS, C. (1980). *Discourse Intonation and Language Teaching* (London).
BROWN, G., CURRIE, K.L. and KENWORTHY, J. (1980). *Questions of Intonation* (London).
CHAFE, W.J. (1970). *Meaning and the Structure of Language* (Chicago).
CHOMSKY, N. (1972). *Studies on Semantics in Generative Grammar* (The Hague).
CHOMSKY, N. and HALLE, M. (1968). *The Sound Pattern of English* (New York).
DIXON, W.J. and BROWN, M.D. (eds.). (1979). *Biomedical Computer Programs,* P-series (Los Angeles).
ENKVIST, N.E. (1979). Marked focus: functions and constraints. In S. Greenbaum, G. Leech and J. Svartvik (eds.), *Studies in English Linguistics for Randolph Quirk* (London), pp. 134-152.
GUSSENHOVEN, C. (in press). Focus, mode and the nucleus. *Journal of Linguistics.*
GUSSENHOVEN, C. (forthcoming). Stress shift and the nucleus.
GUSSENHOVEN, C. and BLOM, J.G. (1978). Perception of prominence by Dutch listeners. *Phonetica,* **35**, 216-230.
HALLIDAY, M.A.K. (1967). Notes on transitivity and theme in English, Part 2. *Journal of Linguistics,* **3**, 199-244.
HALLIDAY, M.A.K. (1970). *A Course in Spoken English: Intonation* (Oxford).
LIBERMAN, M. and PRINCE, A. (1977). On stress and linguistic rhythm. *Linguistic Inquiry,* **8**, 249-336.
NESPOR, M. and VOGEL, I. (in press). Prosodic structure above the word. In A. Cutler and D.R. Ladd (eds.), *Prosody: Models and Measurements* (Heidelberg).
PALMER, F.R. (1974). *The English Verb* (London).
POSTAL, M. (1971). *Cross-over Phenomena* (New York).
SELKIRK, E.O. (1980). *On Prosodic Structure and its Relation to Syntactic Structure* (Bloomington, Indiana).